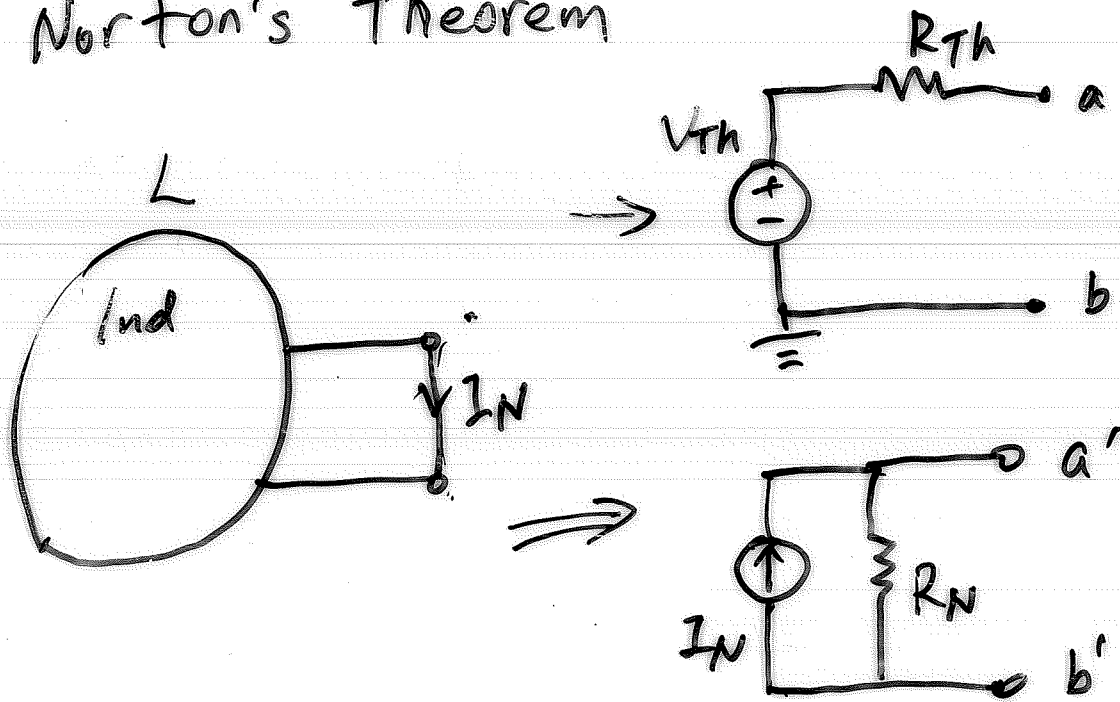
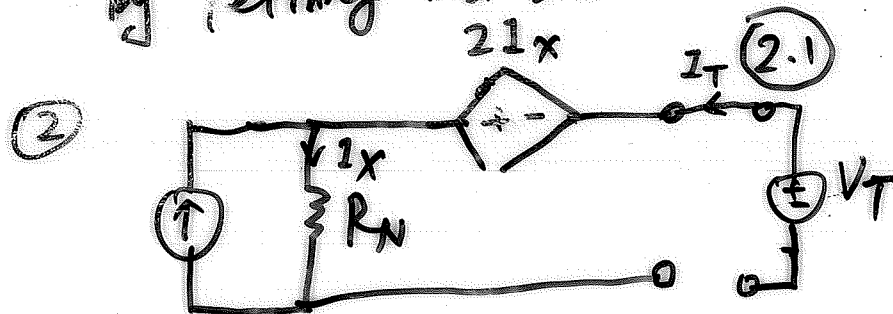


Norton's Theorem



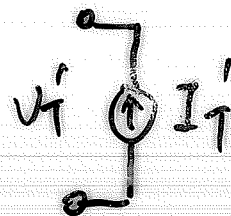
I_N, R_N , $R_N = R_{TH}$

① use approach as we used to get R_{TH} by letting Ind sources $= 0$.



$$R_N = \frac{V_T}{I_T}$$

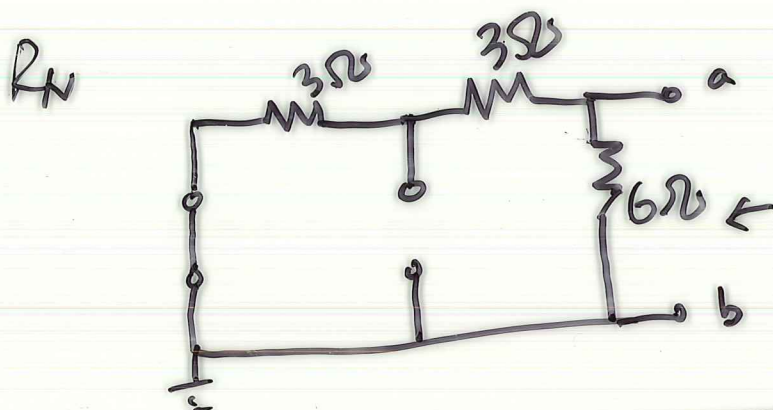
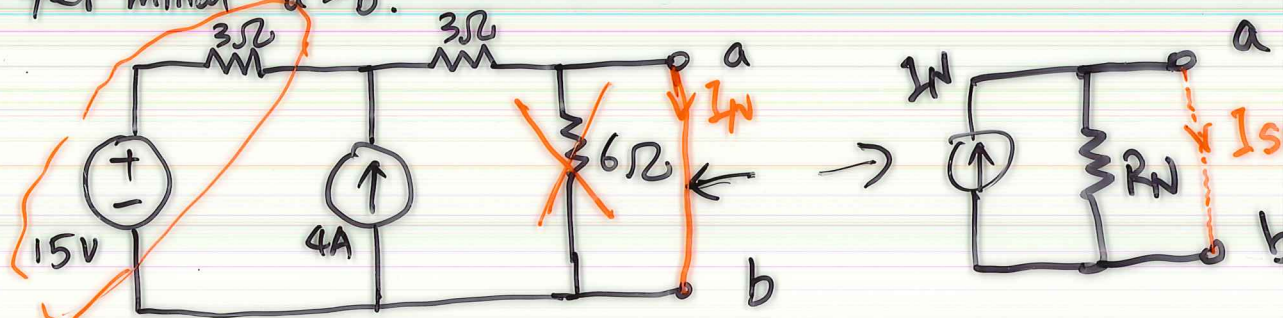
$$R_N' = \frac{V_T'}{I_T'}$$



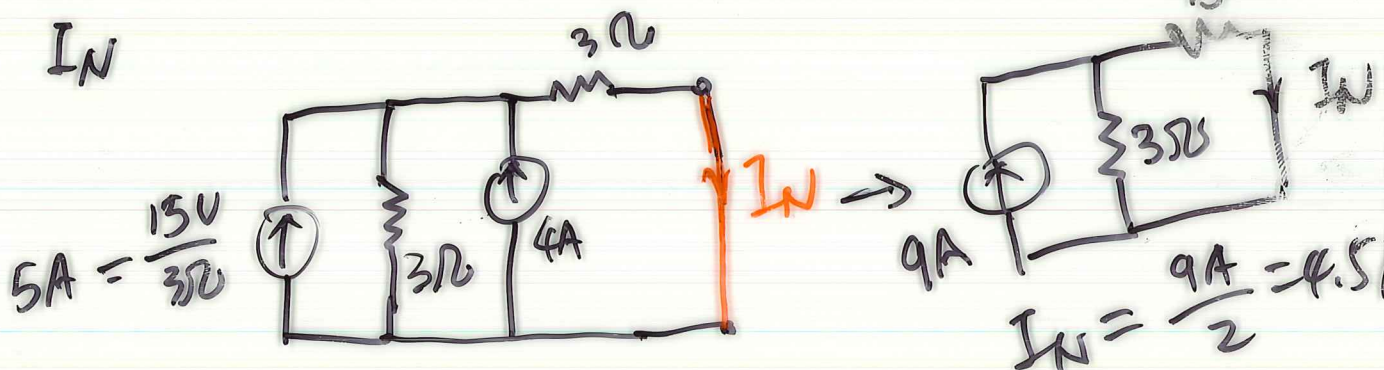
③ $I_N = \frac{V_{TH}}{R_N} \Rightarrow R_N = \frac{V_{TH}}{I_N} \rightarrow V_{oc}$
 $R_N = \frac{V_{oc}}{I_{sc}}$

Example #1

Determine Norton equivalent circuit to the left of terminal a-b.

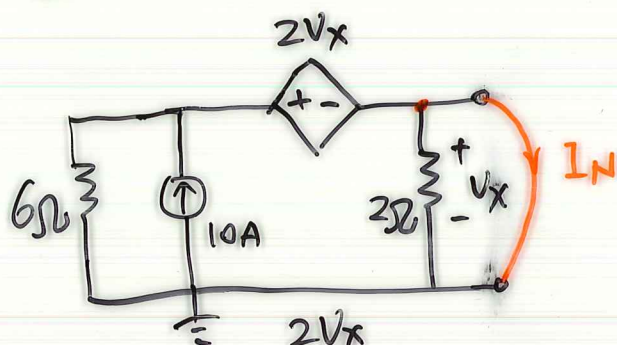
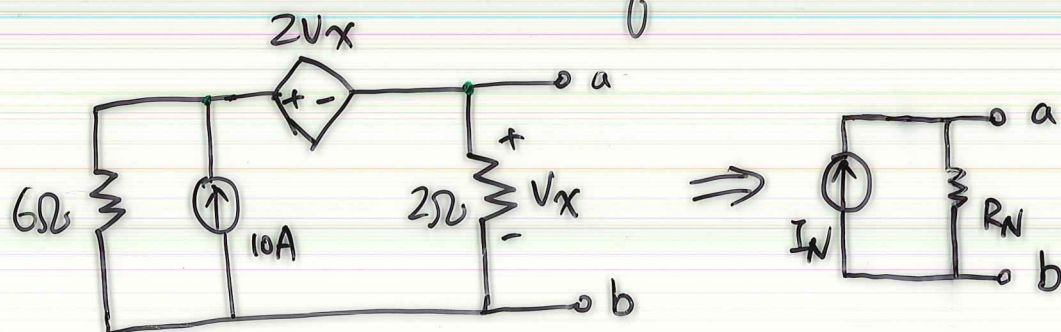


$$R_N = 6\Omega \parallel (3\Omega + 3\Omega) = 3\Omega$$



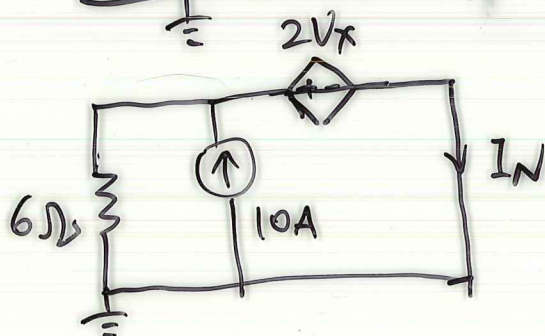
Example #2

Determine the Norton equivalent circuit.



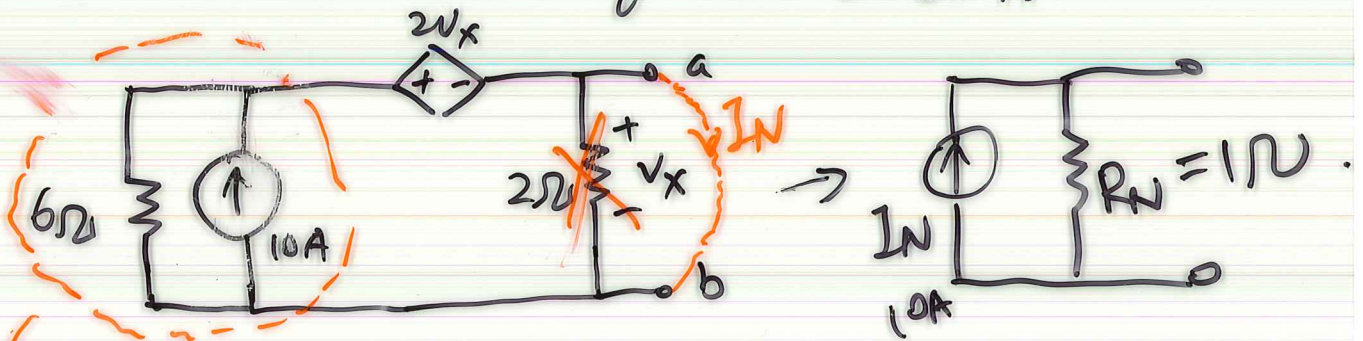
$$V_x = 0$$

$$I_N = 10\text{A}$$

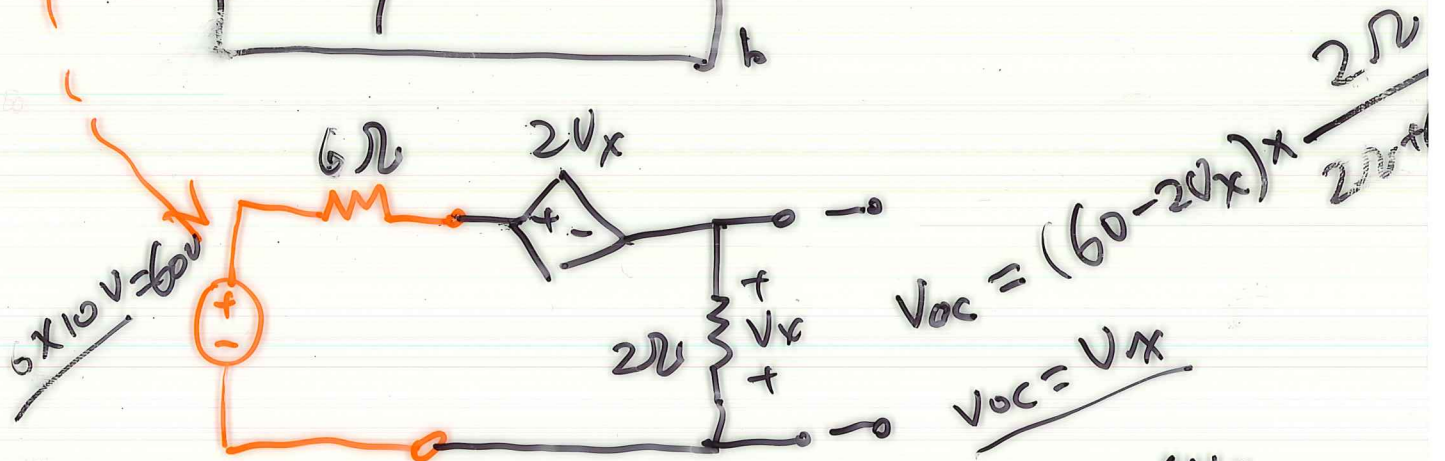
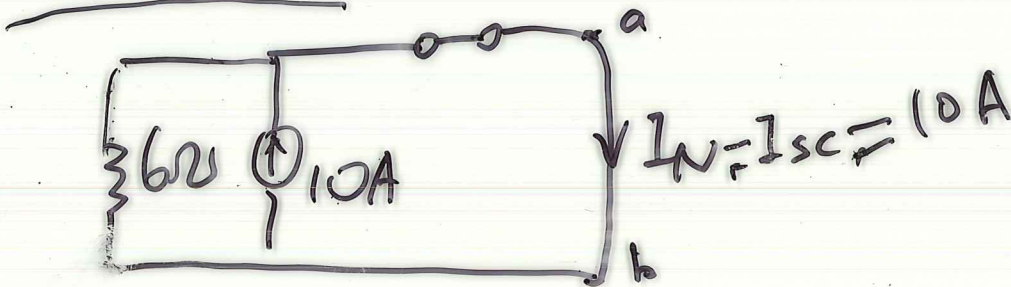


Example # 2

Determine the Norton equivalent circuit.



$$R_N = \frac{V_{oc}}{I_{sc}}, \quad I_{sc} = I_N$$



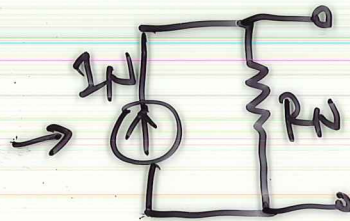
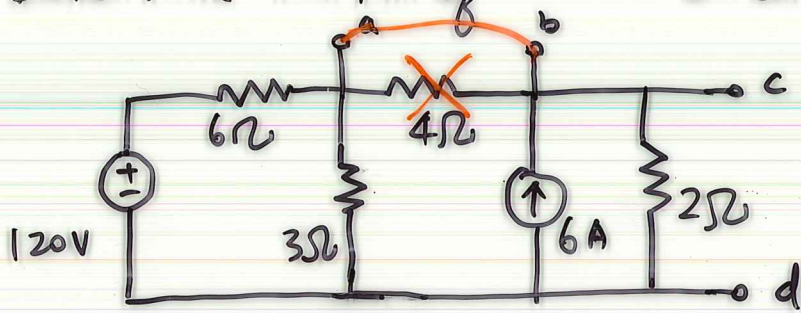
$$(60 - 2V_x) \cdot \frac{1}{4} = V_x, \quad 60 - 2V_x = 4V_x$$

$$V_x = V_{oc} = 10V$$

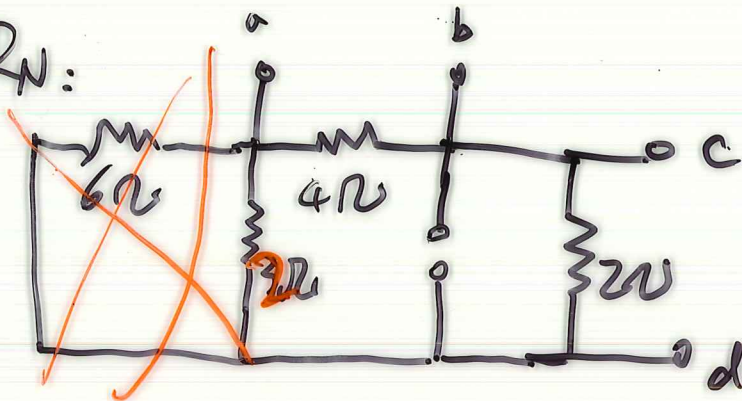
$$R_N = \frac{V_{oc}}{I_{sc}} = \frac{10V}{10A} = 1\Omega$$

Example #3

Determine Norton equivalent circuit between a-b.



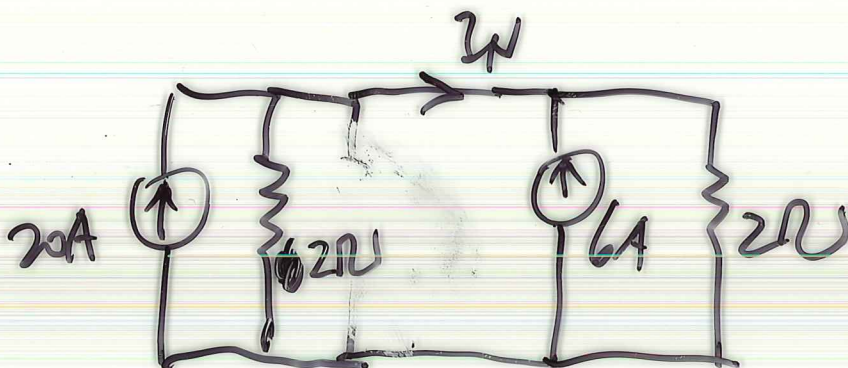
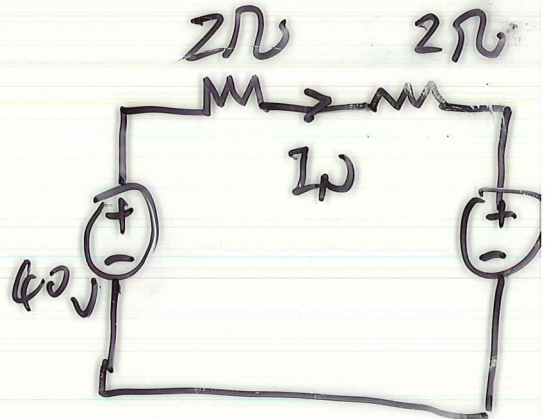
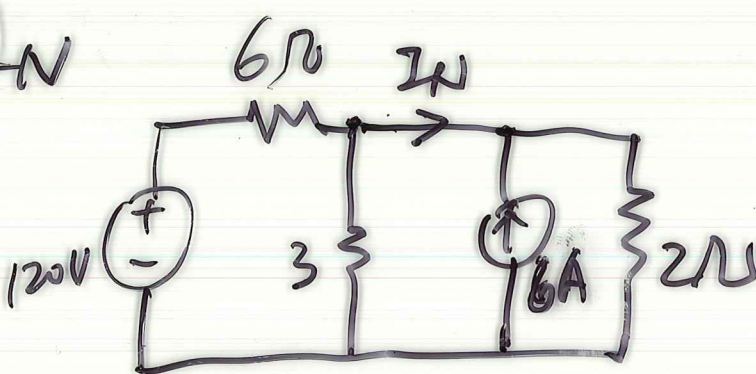
R_N :



$$6 \parallel 3 = \frac{6 \times 3}{6 + 3} = 2\Omega$$

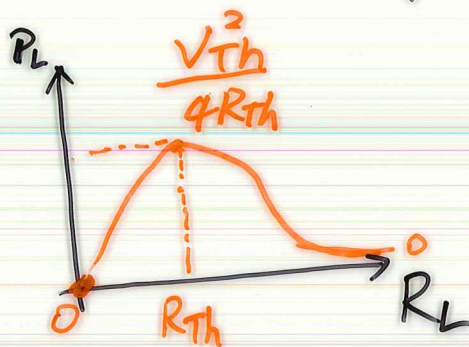
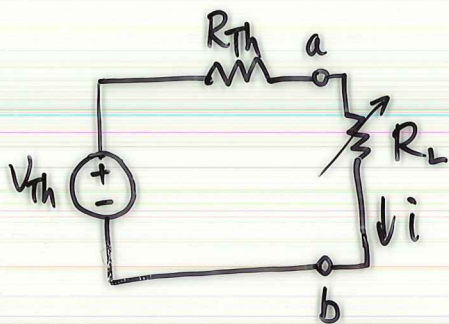
$$R_{ab} = R_N = 4 \parallel (2 + 2) = 2$$

I_N



$$\frac{40 - 12}{4} = I_N =$$

Maximum Power Transfer



$$P_L = i^2 R_L$$

$\swarrow$
0 ∞

$$P_{\max} = \frac{V_{Th}^2}{4R_{Th}}, \text{ maximum power is transferred}$$

$$P_L = i^2 R_L, \quad i = \frac{V_{Th}}{R_{Th} + R_L}$$

$$R_L = R_{Th}$$

$$P_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L$$

$$f(x), \quad f'(x) = 0$$

$$\frac{df(x)}{dx} = 0$$

$$P_L(R_L) = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L, \quad \frac{dP_L(R_L)}{dR_L} = 0$$

$$\frac{dP_L(R_L)}{dR_L} = V_{Th}^2 \times (R_{Th} + R_L)^{-2} \times R_L, \quad (u \times v)' = u \times v' + u' \times v$$

$$(x^n)' = n x^{n-1}$$

$$= V_{Th}^2 \left[-2(R_{Th} + R_L)^{-3} R_L + (R_{Th} + R_L)^{-2} \times 1 \right] = 0$$

$$\frac{-2R_L}{(R_{Th} + R_L)^3} + \frac{1}{(R_{Th} + R_L)^2} = 0, \quad \text{multiply } (R_{Th} + R_L)^3$$

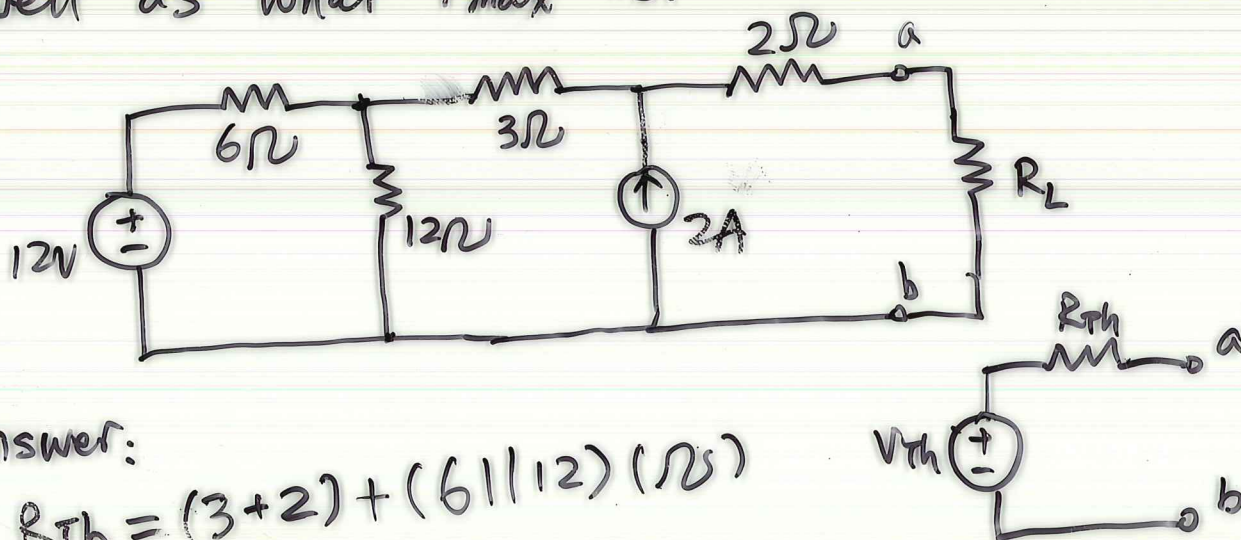
$$-2R_L + R_{Th} + R_L = 0$$

$$\Rightarrow R_L = R_{Th}$$

$$P_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 \times R_L = \frac{V_{Th}^2}{(2R_{Th})^2} \cdot R_{Th} = \frac{V_{Th}^2}{4R_{Th}}$$

Example # 1

Determine R_L for maximum power transfer as well as what P_{max} is.



Answer:

$$R_{Th} = (3+2) + (6 \parallel 12) (\Omega) = 9\Omega$$

$$\textcircled{1} 12V \rightarrow V_{Th}^1 = 12V \times \frac{12}{6+12} = 8V$$

$$\textcircled{2} 2A \rightarrow V_{Th}^2 = 2 \times 3 + (2 \times \frac{6}{6+12}) \times 12 = 14V$$

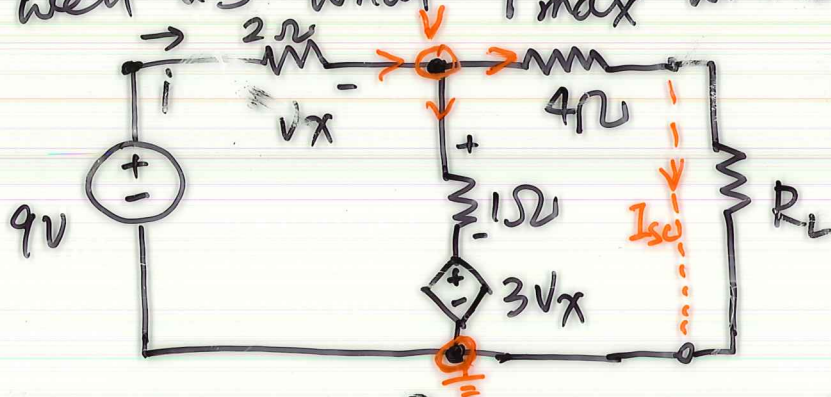
$$\textcircled{3} V_{Th} = V_{Th}^1 + V_{Th}^2 = 22V$$

$$R_L = R_{Th} = 9\Omega$$

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{22^2}{4 \times 9} (W)$$

Example #2

Determine R_L for maximum power transfer as well as what P_{max} will be.



$$V_x = 2i$$

$$V_{15\Omega} = i \times 15 = 5V_x$$

Answer: V_{Th} : $-9 + V_x + 0.5V_x + 3V_x = 0$

$$4.5V_x = 9 \Rightarrow V_x = 2V.$$

$$V_{Th} = 0.5V_x + 3V_x = 7V.$$

$$I_{sc}, R_{Th} = \frac{V_{oc}}{I_{sc}}$$

$$9 - V = V_x$$

$$V + V_x = 9$$

$$12V + 12V_x = 108$$

$$\frac{9 - V}{2} = \frac{V}{4} + \frac{V - 3V_x}{1}$$

$$18 - 2V = V + 4V - 12V_x$$

$$\begin{cases} 7V - 12V_x = 18 \\ 12V + 12V_x = 108 \end{cases}$$

$$19V = 126 \Rightarrow V = \frac{126}{19}$$

$$I_{sc} = \frac{V}{4} = \frac{126}{76} A = 1.66A$$

$$R_{Th} = \frac{7V}{1.66A} = 4.2\Omega$$

$$\Rightarrow$$

$$R_L = R_{Th} = 4.2\Omega$$

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{7^2}{4 \times 4.2} W.$$